HIGH-LEVEL DESIGN (HLD)

1. System Overview

Problem Definition
Customer support systems often fail to provide accurate answers because they do not use internal knowledge effectively. Users receive generic responses, and complex queries require human intervention.

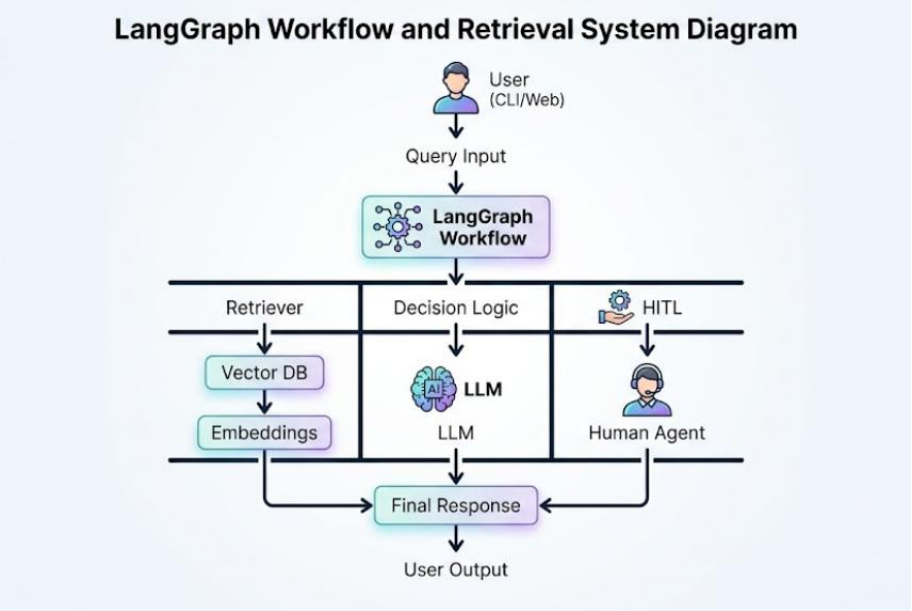
Proposed Solution
A **RAG-based Customer Support Assistant** that:

- Uses a PDF knowledge base
- Retrieves relevant information using embeddings
- Generates contextual answers using an LLM
- Uses a workflow system (LangGraph)
- Escalates queries to humans when needed (HITL)

Scope

- Single PDF input
- Text-based user queries
- Real-time response generation
- Basic escalation system

2. Architecture Diagram



3. Component Description

- **Document Loader**: Loads PDF data
- **Chunking Module**: Splits text into smaller parts

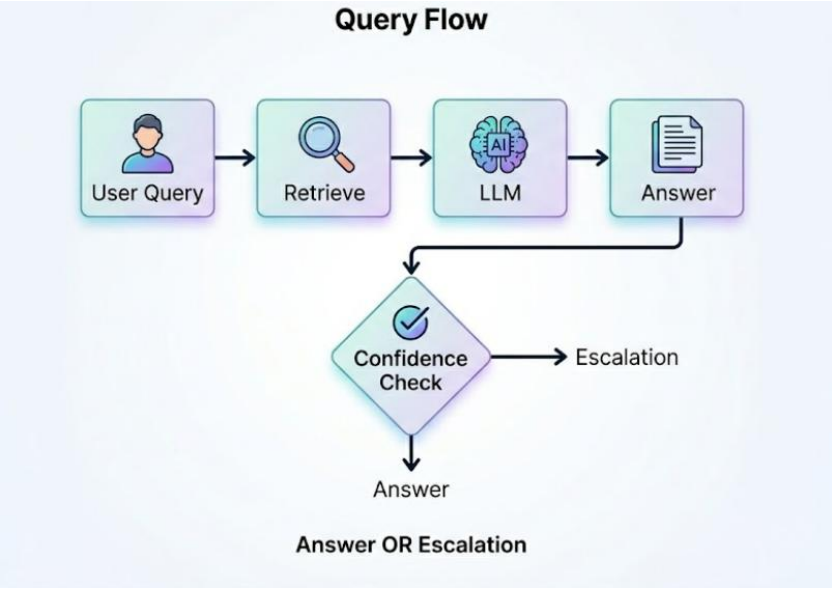
- **Embedding Model**: Converts text into vectors
- **Vector Database (ChromaDB)**: Stores embeddings
- **Retriever**: Fetches relevant chunks
- **LLM**: Generates answers
- **LangGraph Engine**: Controls workflow
- **Routing Layer**: Decides next step
- **HITL Module**: Handles escalation

4. Data Flow

Ingestion Flow

PDF → Text → Chunks → Embeddings → Vector DB

Query Flow



5. Technology Choices

- **ChromaDB**: Lightweight and easy to use
- **LangGraph**: Enables decision-based workflows
- **OpenAI LLM**: Strong text generation
- **Python + LangChain**: Easy integration

6. Scalability

- Large PDFs → better chunking
- High traffic → caching
- Reduce latency → optimize retrieval